
ResearchSynthesisSkill: Individual Report Outline

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Social Network Analysis, Spring 2026
Replace with email or GitHub handle

Abstract

This outline is structured for the required three-page NeurIPS-format individual Skill report. It focuses on `ResearchSynthesisSkill`, the synthesis-side Skill that turns ranked arXiv recommendations into structured daily briefings and selected-paper explanations while preserving explicit evidence boundaries. The final report should explicitly cover functionality, references, results, analysis, and visualization.

1 Functionality

Skill purpose. Define `ResearchSynthesisSkill` as the Skill that explains and summarizes the papers selected by the recommendation-side Skill. It answers: what does each recommended paper contribute, why should the user read it, how do the Top-K papers compare, and what evidence supports each generated claim?

Inputs. List the inputs consumed from the Agent workflow:

- Top-K Recommendation rows with rank, score, paper metadata, evidence source, and ranking rationale.
- Candidate-pool context, `QueryPlan`, `RetrievalQuery`, retrieval metadata, and ranking metadata.
- Optional selected paper, explanation mode, cached full text, or user-provided full text for deep explanation.

Outputs. Describe the outputs: `PaperBriefingItem`, `DailyBriefing`, `PaperDeepExplanation`, evidence-bound claims, reading priorities, Top-K comparisons, trend overview, and a structured evidence boundary. State explicitly that the default daily briefing does not claim PDF or full-text evidence unless full text is actually available for selected-paper explanation.

2 Method and implementation

Per-paper extraction. Explain that `extract_paper(...)` converts each recommendation into a structured `PaperBriefingItem`: summary, contributions, methods, relevance rationale, problem, approach, reading guide, and evidence-bound claims. Provider failures produce deterministic fallback items rather than dropping the paper.

Daily briefing generation. Explain that `generate_briefing(...)` combines extraction results, candidate-pool context, query-plan context, and ranking metadata into an enhanced `DailyBriefing`. The briefing contains an executive summary, summary table, highlighted paper, paper-level items, trend overview, Top-K comparisons, reading priorities, and evidence boundary.

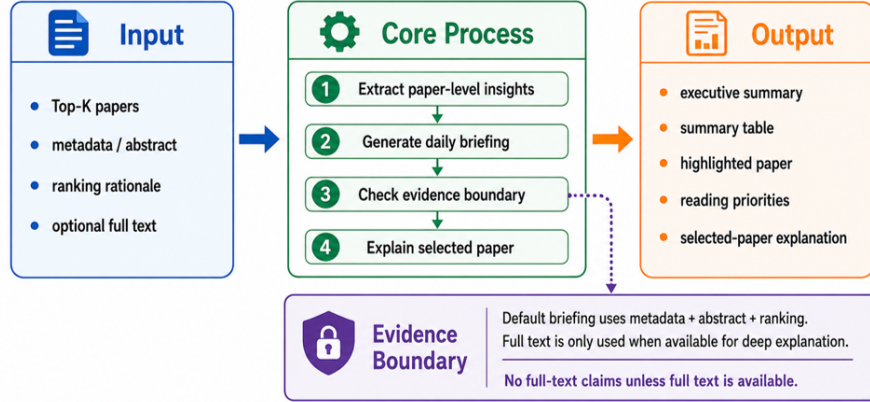


Figure 1: Planned individual-report visualization: ResearchSynthesisSkill consumes ranked recommendations and produces paper extraction items, enhanced daily briefings, and selected-paper deep explanations with evidence labels.

Table 1: Planned ResearchSynthesisSkill results table.

Evaluation	Metric	Fixture-backed result
Daily briefing	required-section coverage	1.0
Daily briefing	Top-K coverage	1.0
Daily briefing	evidence boundary present	yes
Daily briefing	forbidden full-text claims	none in passing run
Deep explanation	method-mode completeness	0.86
Fallback behavior	malformed or missing evidence	structured result

Selected-paper deep explanation. Explain that `explain_paper(...)` supports method, experiment, and limitations modes. It prefers full text when supplied or cached; otherwise it falls back to abstract, then metadata. Each fallback is explicitly labeled so the user can distinguish full-paper analysis from abstract-only or metadata-only output.

3 References and related work

Use this section to position the Skill relative to evidence-grounded summarization:

- Retrieval-augmented generation motivates separating retrieval evidence from generated synthesis (Lewis et al., 2020).
- Faithfulness work in summarization motivates evidence labels and forbidden full-text claims (Maynez et al., 2020).
- Hallucination surveys motivate explicit abstentions and source boundaries when evidence is missing (Ji et al., 2023).
- arXiv metadata and abstracts define the default evidence available to the briefing (arXiv, 2026).

4 Results

Report deterministic offline results from the enhanced briefing and explanation evaluators. The final report should state that the evaluation checks structural completeness, evidence discipline, and mode-specific coverage rather than subjective writing quality.

Qualitative demo results. Use the presentation example: after the recommendation Skill selects papers, this Skill generates a daily briefing with an executive summary, summary table, per-paper

items, reading priorities, and evidence boundary. For a selected paper, it can explain the method, experiment, or limitations rather than producing one generic summary.

5 Analysis

What works and why. Argue that the Skill improves usefulness because it turns Top-K rows into reading decisions. Evidence labels reduce hallucination risk, Top-K comparisons make recommendations easier to compare, reading priorities help users decide what to read first, and mode-specific explanations expose missing information instead of hiding it inside a generic paragraph.

Ablations and error analysis. Include these analyses in the final report:

- Enhanced briefing versus generic summary: the evaluator fails generic text even when sections exist.
- Default briefing with and without forbidden full-text claims: PDF/full-text claims are rejected unless full-text evidence was used.
- Full-text explanation versus abstract/metadata fallback.
- Explanation completeness by mode, with the 0.86 method-mode result showing one intentionally missing required field.

Limitations. State the limits plainly: default daily briefings are based on metadata, abstracts, ranking evidence, and candidate-pool context; full-text analysis is only for selected-paper explanation; fake-provider results test structure rather than human preference; and current explanation completeness does not measure factual correctness beyond available evidence labels.

6 Page allocation for the final three-page report

Table 2: Suggested three-page allocation. References do not count toward the page limit.

Page target	Content
Page 1	Functionality, inputs/outputs, Figure 1
Page 2	Method, evidence boundary, related work
Page 3	Results, ablations, limitations, StudyClawHub metadata

Checklist before finalizing

- Replace author placeholders and add the published StudyClawHub Skill URL.
- Keep the final individual report at three pages before references.
- Include at least one visualization; Figure 1 satisfies the requirement.
- Do not present abstract-only or metadata-only output as full-paper analysis.

References

- arXiv. arXiv API User Manual. <https://info.arxiv.org/help/api/index.html>.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuettler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. *ACL*, 2020.